

The role of the hippocampus in tracking personal goals

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Abstract—The hippocampus plays a central role in both episodic memory and spatial navigation, and recent research has shown that these cognitive processes share overlapping neural mechanisms. In particular, Deuker et al. demonstrated that the hippocampus may encode a spatio-temporal event map, in which neural activation patterns elicited by specific stimuli are correlated with the remembered spatial locations of those stimuli [1]. The present study followed a design closely modeled on that of Deuker et al., with several additional elements. Importantly, our design allows the comparison of encoded stimuli with newly presented stimuli, extending the analysis beyond Deuker et al.’s focus on encoded stimuli alone. Although we were unable to fully reproduce Deuker et al.’s findings, our results reveal related patterns while also questioning the mnemonic aspect of the picture viewing task, a crucial aspect in the retrieval of the spatio-temporal event map.

I. INTRODUCTION

The hippocampus has long been recognized as a central structure in the brain’s memory system, particularly for its role in episodic memory, the recollection of personally experienced events [2]. Beyond its mnemonic function, extensive animal studies have revealed that the hippocampus is also crucial for spatial navigation. The discovery of place cells—neurons that fire when an animal occupies specific locations in its environment, provided the first evidence of a spatial coding mechanism in the brain [3]. Building on these discoveries, researchers have also identified cells within hippocampal neurons that exhibit activity patterns corresponding to distinct moments within a temporal interval, suggesting a neural representation of time comparable to that of space [4]. Although similar findings have been supported in humans [5], [6], achieving this level of neural specificity remains challenging due to the invasiveness of the required recording methods. Recent research has proposed a possible integration of these three dimensions, suggesting that the hippocampus may encode memories in a spatiotemporal manner to facilitate more efficient retrieval [1]. Functional MRI (fMRI), long used to study memory processes in humans [7], [8], provides a non-invasive means to investigate this spatiotemporal organization. In addition to the spatial and temporal dimensions, the relevance of memories has also been shown to influence their stability. Montagnin et al. recently demonstrated that this relevance might be temporally encoded within the hippocampus, with distinct activations in the left posterior hippocampus for current goals and in the left anterior hippocampus for past or future goals [9].

The present study focuses on the subjective spatial organization of memory within the hippocampus and discusses the various metrics that can be applied in fMRI analyses to investigate how the brain encodes such information.

II. EXPERIMENTAL SETUP

The experiment analyzed in this study is divided into five major phases. In the first phase, the **familiarization task**, participants learn the spatial organization of a (VR)-like island environment by navigating all its possible paths. The second phase, the **pre-viewing task**, consists of six repetitions of the presentation of 64 unique images. These images are categorized into *maya* (11), *pirate* (10), *viking* (11), *wealth* (11), *food* (11), *drink* (10), *faces* (32), and *objects* (32), as well as into *old* (48) and *new* (16). The latter categorization is relevant for the third phase, the **encoding task**.

In this phase, participants are asked to walk to several checkpoints with a given mission, for example, “Viking–Drink.” At each checkpoint, they are presented with an image and asked to indicate whether it is goal-relevant or not. Only images categorized as *old* are shown during the encoding process. The encoding task occurs three times, three weeks apart, and each session consists of two runs.

The fourth phase, the **post-viewing task**, is a repetition of the pre-viewing task and takes place one day after the last encoding session. Finally, the last phase, the **memory task**, occurs on the same day and consists of a questionnaire on 64 images: the 48 categorized as *old* and seen during the encoding phase, and 16 entirely new images. Participants are asked to indicate whether they have seen each image before, to place a marker on the island indicating where they remember having seen the image (in the water if never seen), and to specify when they remember seeing it: never, yesterday, one week ago, or two weeks ago. Additionally, their confidence for both their spatial and temporal responses was collected on a scale from 1 to 100.

Only phases 2, 3, and 4 were performed in the scanner, and this study focuses on the analysis of data from the pre- and post-picture viewing tasks.

III. METHODS

A. Dataset and Preprocessing

The final dataset consists of fMRI runs from 28 participants, all recruited from the University of Geneva. The data were acquired on a 3T scanner and preprocessed using *fMRIPrep*

with the following steps: brain extraction, brain tissue segmentation, spatial normalization, head-motion estimation, slice-timing correction, susceptibility distortion correction (SDC), registration to the anatomical scan, and normalization to standard space. Smoothing (5 mm FWHM) was applied for whole-brain analyses but not at the hippocampal level.

B. GLM

All GLM analyses in this study are performed using the `nilearn.glm` package. The design matrices include the following confounders: `trans_x`, `trans_y`, `trans_z`, `rot_x`, `rot_y`, `rot_z`, `global_signal`, `csf`, `white_matter`, as well as both the first and second derivatives of the trial regressors. The hemodynamic response function (HRF) is modeled using the SPM approach. The fMRI runs from both the pre- and post-viewing tasks are concatenated and separate regressors are included for each run to preserve their temporal structure, enabling the simultaneous modeling of all trials within a unified design matrix. This allows all contrast to be computed per session (pre and post) but also across sessions which allows to observe whether a contrast is stronger in the post session than in the pre session. A gray matter mask derived from the MNI152 template is used, resampled to each participant's fMRI space, and an additional hippocampal mask from the Harvard-Oxford atlas is used for the RSA analysis. A group mask for group-level analyses is computed by retaining only voxels present in at least 80% of the participants.

C. Representational Similarity Analysis

Representational Similarity Analysis (RSA) is a method that allows studying the correlation between neuronal representations and model representations, in this case, the spatial distance between stimuli. Several metrics can be computed to capture these representations [10]. In this study, Pearson's correlation was used to compute the neuronal similarity matrix, while the final analysis of the similarity between the neuronal and model representations was performed using Spearman's correlation, following common practice and the approach of Deuker et al.

1) *Model Matrix*: The model matrix represents the idealized representation to be studied and, in this case, is built from the positions estimated by the participants. For all participants, an initial selection includes only images correctly classified as previously seen. Images placed in the water (classified as not seen) or images placed on the island but never actually presented during the encoding sessions are excluded, as they are not relevant for studying the participant's subjective spatial representation. The number of resulting images ranges from 17 (35.31%) to 48 (100%), with a mean of 33 (68.75%) images. Finally, the Euclidean distance between each image's judged position (X, Y, Z) is computed.

2) *Neuronal Matrix*: The neuronal matrix is defined as a correlation matrix comparing the neural response to a specific image with the neural response to another image. Its computation begins with a subject-level GLM in which a distinct

regressor is assigned to each of the 384 trials (6 repetitions of 64 images). Beta maps are extracted for each trial, flattened, and then Pearson-correlated with every other trial across all repetitions, following the method of Deuker et al. The mean correlation across repetitions is then computed, and only trials present in the model matrix are retained.

3) *Low and High Distance Split*: A common complementary method to study the representation of space in the hippocampus is to divide the neuronal correlation values by their relative distance in the model matrix. This was done by computing the median distance per participant and separating all values of the neuronal matrix into low and high distances.

4) *Neuronal Matrix Metric*: While it is quite common to use Pearson's correlation as a similarity measure or a dissimilarity measure (1 - Pearson), it is quite debatable to consider it as a linear scale of gradual progression from dissimilar to similar. Indeed, Pearson's correlation scale goes from anti-correlated to uncorrelated to correlated, with the flow of information being not gradually similar but gradually random from -1 to 0, and gradually correlated again from 0 to 1. Comparing a neuronal matrix of Pearson coefficients to a model matrix of Euclidean distance by Spearman correlation can thus negate this intrinsic effect of correlation. In order to study whether there is indeed a distinction between correlation and similarity, the positive and negative Pearson correlations were separated by building two neuronal matrices: one with values ranging from -1 to 0 (all values > 0 set to 0, negative correlation effect) and one with values ranging from 0 to 1 (all values < 0 set to 0, positive correlation effect). This method allows building two matrices with a clear linear increase effect.

IV. RESULTS

A. Control

Before pursuing the analysis of the spatial encoding of memory in the hippocampus several control analysis are done to ensure good GLM procedure and preprocessing pipeline.

1) *Faces vs. Objects*: The Fusiform Face Area (FFA) is a well-established region of the brain specialized for face processing and is therefore commonly used as a reference in GLM-based fMRI studies [11]. To assess activation related to face perception, a contrast of faces (Maya, Pirate, Viking) versus objects (Wealth, Drink, Food) was computed separately for the pre- and post-sessions. As shown in Figure 1 and Table I, after FDR correction at $\alpha = 0.05$, the group-level results revealed clear bilateral activation in the FFA, along with strong activation in the visual cortex for both sessions.

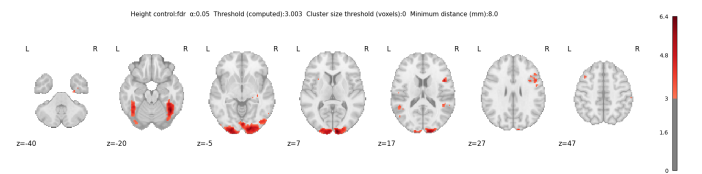


Fig. 1. Group-level *Faces > Objects* Z-map with FDR correction at $\alpha = 0.05$ for the post-session.

Session	Cluster	x	y	z	Peak Stat	Size (mm3)
Pre	1	16.5	-99.5	-0.5	6.97	22171
	2	-13.5	-104.5	14.5	6.33	10390
	3	16.5	-7.0	-15.5	4.83	468
	4	-43.5	-44.5	-23.0	4.62	1140
	5	4.0	48.0	-23.0	4.31	1328
Post	1	11.5	-102.0	7.0	6.41	26093
	2	-23.5	-97.0	-3.0	6.13	13937
	3	46.5	10.5	17.0	5.02	1500
	4	21.5	-4.5	-15.5	4.66	312
	5	-21.0	-4.5	-15.5	4.40	328

TABLE I

5 AREAS OF ACTIVATION FOR FACES > OBJECTS CONTRAST IN PRE- AND POST-SESSION GLM ANALYSIS WITH THE HIGHEST PEAK Z STATISTIC. BILATERAL FFA REGION HIGHLIGHTED IN BOTH RUNS.

2) *Old vs. New*: After verifying that the fMRI preprocessing and GLM estimation appeared satisfactory, we examined whether the brain responses to old versus new images were distinct in order to control for significant memory encoding. However, at the group level, no cluster survived FDR correction at $\alpha = 0.05$ for any of the pre-, post-, or post- vs. pre-sessions. Additional GLMs were thus computed with or without smoothing and at both the whole-brain and hippocampus levels. None yielded significant results. As shown in Figure 2, the subject-level results vary, with significance observed in all types of comparison (pre-, post-, and post- vs. pre-sessions) and a relatively equal number of subjects with at least one significant region after FDR correction for both pre- and post-sessions.

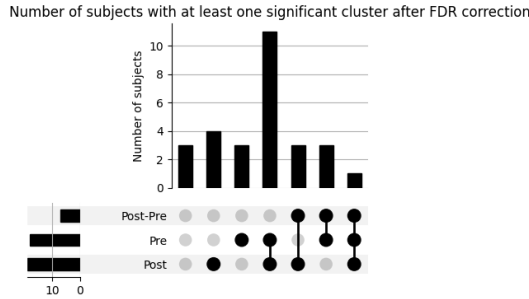


Fig. 2. Result of Old vs. New contrast GLM analysis at the subject-level. Number of subjects with at least one significant cluster after FDR correction for all combinations of pre, post and pre-post sessions.

3) *Random Contrast*: In order to test whether the effect observed in the old vs. new contrast at the subject level is truly random, a random-contrast GLM was built by creating two randomly balanced groups of images. Similar to the old vs. new contrast, no cluster survived FDR correction; however, Figure 3 shows that the distribution of significant clusters appears to be skewed toward the post-session.

B. Hippocampal Spatial Representation

Although the results so far do not show evidence of significant memory encoding effects, no control was performed by Deuker et al., and they still observed significant hippocampal spatial representation. It was therefore decided to attempt to reproduce their RSA results.

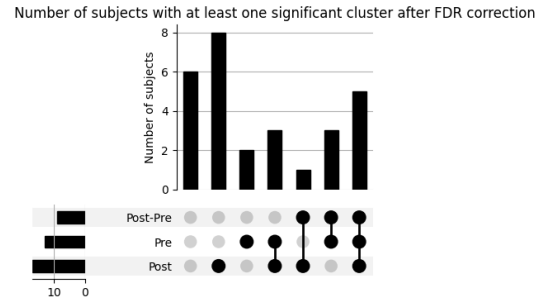


Fig. 3. Result of random contrast GLM analysis at the subject-level. Number of subjects with at least one significant cluster after FDR correction for all combinations of pre, post and pre-post sessions.

1) *RSA*: The Representational Similarity Analysis yields a negative spearman correlation coefficient of -0.0136 ± 0.0101 . This correlation is however not significant at the group-level with an uncorrected p-value of 0.1496. Further analysis shows that while images that are close in space (low distance) seem to be positively correlated, this is also the case for images far in space (high distance) (Figure 4).

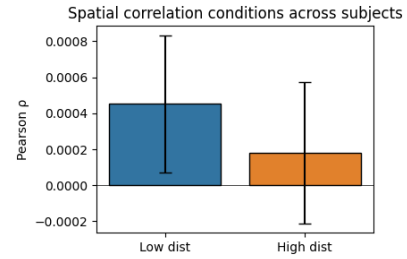


Fig. 4. Median split of the Pearson's correlation values depending on their corresponding distance in the model matrix. The average at the group level for low and high distance is shown $\pm$ SEM

2) *Pearson's distinct effects RSA*: Interestingly, the results show no distinction between positive and negative effects, with a negative Spearman correlation for both (Figure 5). This can be understood as two distant objects having a lower correlation coefficient than two close objects, regardless of the sign distinction.

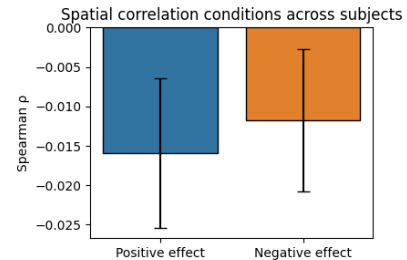


Fig. 5. Distinct Representational Similarity Analysis of positive and negative Pearson's correlation. The average Spearman correlation across participant is shown $\pm$ SEM.

3) *Judged Position Confidence*: In addition to the judged location of an image (X, Y, Z), a confidence rating between

1 and 100 regarding this placement was obtained from the participant. This confidence was studied to assess its possible correlation with the strength of memory for a given image. With a Pearson correlation of 0.0152 and a p-value of 0.6004, the confidence appears not to be correlated with the spatial accuracy of the participants. It is thus difficult to define a minimum threshold of confidence for stimulus selection (Figure 6). An additional RSA was, however, still performed using only trials with confidence higher than the participant’s median, ensuring data were available for all participants. This, however, yielded no significant results.

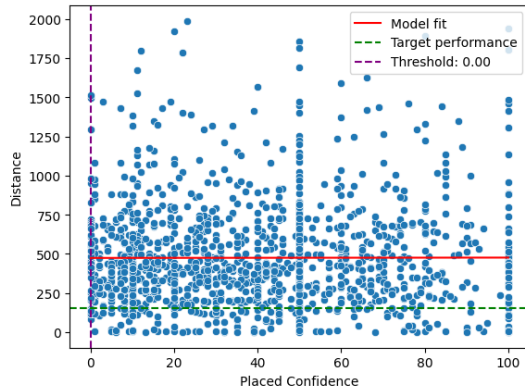


Fig. 6. The distance from the judged position to the true position of the stimuli is calculated and plotted against the confidence of the judgment. The linear regression model is shown in red, and the target performance, defined as the average distance minus the standard deviation, is shown in green. In purple is shown the factual minimum confidence threshold required to meet the target performance.

C. Discussion

The present study aimed to reproduce the findings of Deuker et al., who reported in their paper *An event map of memory space in the hippocampus*, that the bilateral hippocampal response to two distinct stimuli was significantly negatively correlated with the remembered distance separating them. This result supports the hypothesis that the hippocampus encodes memories differently depending on their spatial organization [12]. In our study, while we obtained a comparable correlation coefficient from the RSA of judgment distances and neural responses, none of the results reached statistical significance.

Several factors may account for this discrepancy. First, important distinctions in experimental design need to be noted. The distance measure used by Deuker et al. in their RSA was not derived from estimated positions but directly obtained from participants’ reports. This proxy used in our design may have weakened the correlation between the calculated distances and the neural responses. In fact, for n different stimuli, our design provides only n position values, whereas Deuker et al. obtained $n(n-1)$ distance values for each pair of stimuli.

In addition, our design presented participants with 48 different stimuli, compared to 16 in Deuker et al. The fact that three times more stimuli were used may have made memorization more difficult. Another key difference lies in

the picture-viewing task from which neural activations were obtained. While our design involved six identical repetitions of 64 stimuli, Deuker et al. used a fully randomized trial order in which 16 stimuli were presented 12 times each. This ordering effect in our design may have introduced noise related to working memory or other associated processes. Moreover, the smaller number of repetitions in our study could have made the final activation per stimulus less robust, potentially explaining the lower statistical significance of our results.

Most importantly, Deuker et al. restricted their analysis to only old stimuli, i.e., those seen during the encoding sessions. They did not report a specific activation in the bilateral hippocampus for these images compared to images that had never been seen before. In fact, as we showed in this study, no clear distinction between old and new stimuli could be observed. This result may seem unexpected, as memory tasks have long been known to elicit hippocampal activation. However, the fMRI data used here were not obtained from a memory task, but from a picture-viewing task. This distinction is important: participants were never explicitly asked to engage in any form of memory retrieval. We only hypothesized that some degree of mnemonic processing might occur spontaneously as they viewed the images. This consideration also raises questions about the mnemonic interpretation of Deuker et al.’s findings, as a picture-viewing task may not elicit strong memory-related activation, even at the hippocampal level.

To conclude, the present study challenges the mnemonic interpretation of Deuker et al.’s findings by showing that our picture-viewing paradigm does not produce a strong difference in hippocampal activation between old and new images. Nevertheless, these findings should be interpreted with caution, as various other differences in experimental design and methodological factors could also explain the absence of significant effects in this study.

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CODE

The code and analysis scripts for this project were done in python and are available at:

github.com/oskarboesch/island_semester_project.git

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